Final Project Report

Group 17

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DTSC301 • Machine Learning for Data Science-1 • December 13, 2025

Abstract

This project aims to build a reliable framework for predicting the market value of professional football players using the FIFA dataset. Because player valuation depends on a wide range of factors, including technical skills, physical attributes, positional responsibilities, and contract-related variables, it represents a multidimensional and challenging prediction problem. To handle this, the study uses a full machine-learning workflow that includes data exploration, feature engineering, feature selection based on correlation and PCA, and testing six different regression models.

The final version of the workflow is stronger than the earlier one, which had issues with inconsistent feature selection and weak explanations for some preprocessing steps. The updated pipeline now handles missing data more carefully, uses domain knowledge to engineer better features, and applies a clearer approach to reducing dimensions.

Our six models include Linear Regression, Lasso Regression, Decision Tree, Random Forest, Gradient Boosting, and Support Vector Regression, which were assessed using MAE, RMSE, R^2, Adjusted R^2, and model latency. The results show that the ensemble models, especially Random Forest and Gradient Boosting, perform the best while still being efficient to run. Overall, the study also points out which factors are most strongly linked to a player's market value and explains the model behaviour in a clear, sports analytics context.

1 Introduction and Problem Statement

Player valuation has become extremely important in modern football. Clubs rely on these valuations to make decisions about recruitment, contracts, and long-term planning. As the transfer market becomes more competitive, teams are turning to analytical tools to estimate a player's worth more accurately. Machine-learning models help combine many types of information, such as technical skills, physical traits, playing position, and contract details, to produce a more reliable market value.

This project builds a machine-learning system that predicts a player's market value using the FIFA dataset. Since the dataset includes detailed attributes for thousands of players, it provides a strong base for understanding the factors that shape player value. The project follows a complete workflow: data cleaning, exploratory analysis, feature engineering, feature selection, and testing six different regression models.

Compared to the interim version, the final design is more structured and justified. It handles missing or inconsistent data more carefully and uses a clearer feature-selection approach that combines correlation analysis with PCA. These improvements make the modelling process more stable, interpretable, and accurate. In the end, the project not only compares how well different models perform but also highlights the player attributes that most strongly influence market value and explains why.

2 Dataset and Data Access

The dataset used in this project is the FIFA Player Dataset from Kaggle, which provides detailed information on about 18,000 professional footballers. Each record includes demographic details such as age, height, and nationality, physical attributes such as height and weight, skill ratings such as overall and potential, contract information, and positional data. Because it contains many different types of player characteristics, the dataset is well suited for predicting market value, which is influenced by several interconnected factors.

During the first review, the dataset was checked for missing values, duplicate rows, inconsistent category labels, and skewed distributions, especially for the target variable (value_eur). The interim report pointed out that missing values and messy categorical entries required a more organised approach. In the final version, the process was improved by standardising missing-value indicators, removing columns with no useful information, and using median or mode imputation depending on whether a feature was numerical or categorical.

The exploratory analysis showed clear variation in player value across positions, ratings, and contract duration. This confirmed the need for careful feature engineering. Statistical summaries and visual tools such as histograms, scatterplots, and correlation heatmaps were used to identify early relationships in the data. One of the strongest patterns was the positive relationship between market value and attributes like overall rating and potential. These insights helped shape later decisions about feature selection and model design.

Overall, understanding the structure of the dataset influenced the feature-engineering process and addressed several issues from the interim submission. This stage improved the consistency of the dataset and ensured that the final machine-learning pipeline was built on a clean and well-prepared foundation.

FIFA Player Dataset Drive Link- https://drive.google.com/file/d/108abZkoGca5V10qo7Eglgp5mSS0qNuq/view?usp=sharing

3 Code and Implementation

The implementation of the player valuation prediction project was structured to ensure modularity, reproducibility, and clarity. The codebase is organized into separate scripts and notebooks, each dedicated to a specific stage of the workflow: data preprocessing, feature engineering, feature selection, model training, evaluation, and visualization.

3.1 Environment and Dependencies:

The project was implemented using Python 3.10 in a Jupyter Notebook environment. Key dependencies include `pandas` and `numpy` for data manipulation, `scikit-learn` for preprocessing, feature selection, and machine learning models, and `matplotlib` and `seaborn` for visualizations. Additional libraries, such as `joblib` and `pickle`, were used for model and dataset serialization. The environment setup and installation of dependencies are fully documented in the README file, ensuring reproducibility.

3.2 Implementation Details:

- **Data Cleaning and Preprocessing:** Scripts handle missing values, duplicates, and outlier checks, standardizing the dataset before modelling.
- **Feature Engineering:** Derived features such as Remaining Contract Years, BMI, and Potential Growth were generated through dedicated functions, making the workflow reproducible and easily extendable.
- **Feature Selection:** Correlation-based filtering and PCA-based selection were implemented in modular scripts, allowing dynamic adjustment of thresholds and explained variance

parameters.

- **Encoding and Scaling:** Categorical features were processed using label and one-hot encoding as appropriate, while numerical features were standardized to ensure model compatibility.

- **Model Training and Evaluation:** Six regression models were trained on the processed dataset. Each model was implemented in a separate script for clarity, with hyperparameters configurable through a centralized configuration file. Evaluation metrics including MAE, RMSE, R^2, and Adjusted R^2 were calculated for both training and test sets.

3.3 Deviations and Additional Techniques:

While the initial plan focused primarily on linear and tree-based models, the final implementation included Support Vector Regression to capture non-linear relationships in player valuation. Hyperparameter tuning was performed using cross-validation to enhance model performance. Intermediate data outputs and feature importance scores were saved for easier inspection and reproducibility.

All instructions to run the code, including environment setup, data paths, and execution order, are provided in the README file accompanying the repository. This ensures that the workflow can be replicated or extended without ambiguity.

4 Experiments and Results

The performance of all six models was assessed using four error-based metrics—Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2, and Adjusted R^2—along with latency to measure computational efficiency. Table 1 summarises the results.

4.1 Overall Performance

Tree-based ensemble models, namely Random Forest and Gradient Boosting, performed the best across most evaluation metrics. In contrast, the Support Vector Regressor (SVR) performed the worst, showing the highest error values and a negative R^2 score, indicating poor predictive capability.

4.2 Linear Models: Linear Regression and Lasso Regression

Linear Regression and Lasso Regression produced identical results (MAE $\approx 3.30 \times 10^5$, RMSE $\approx 1.03 \times 10^6$, $R^2 \approx 0.9625$). This suggests two important points. First, multicollinearity among the selected features was already low. Second, the feature-selection pipeline successfully removed many redundant predictors, leaving little scope for the Lasso penalty to further improve performance.

Both linear models were also extremely fast, with inference latency under 0.005 seconds, making them computationally efficient despite their lower predictive power compared to ensemble methods.

4.3 Decision Tree Regressor

The Decision Tree Regressor achieved moderate accuracy (MAE $\approx 1.71 \times 10^5$, RMSE $\approx 6.69 \times 10^5$). While decision trees are capable of capturing non-linear patterns, they are prone to overfitting when used in isolation. This explains the higher error values compared to ensemble models and the slightly lower R^2 score (approximately 0.984).

Despite this, the model demonstrated very low latency (approximately 0.003 seconds), reflecting the computational simplicity of a single-tree structure.

4.4 Random Forest Regressor

The Random Forest Regressor delivered some of the strongest results among all models. It achieved the lowest MAE (approximately 1.43×10^5), a strong RMSE (approximately 7.02×10^5), and a high R^2 score (approximately 0.9826). These results highlight the benefit of averaging predictions across multiple trees, which reduces variance and improves generalization.

However, the model's latency was higher (approximately 0.017 seconds), which is expected given the computational cost of evaluating a large ensemble of trees.

4.5 Gradient Boosting Regressor

Gradient Boosting performed comparably well, achieving an MAE of approximately 2.21×10^5, RMSE of approximately 6.67×10^5, and an R^2 score of approximately 0.984. While its MAE was slightly higher than that of Random Forest, it marginally outperformed Random Forest in terms of RMSE.

This behaviour is consistent with the nature of boosting algorithms, which focus on correcting residual errors at each stage. The model also exhibited moderate latency (approximately 0.0068 seconds), lower than Random Forest due to the use of fewer trees.

4.6 Support Vector Regressor (SVR)

The Support Vector Regressor performed the worst among all evaluated models. It recorded very high error values (MAE $\approx 1.88 \times 10^6$, RMSE $\approx 5.55 \times 10^6$) and a negative R^2 score (approximately -0.087). A negative R^2 indicates that the model performs worse than a naive baseline that predicts the mean of the target variable.

SVR likely struggled due to the high dimensionality of the dataset after categorical encoding, the presence of strong non-linear interactions, and its sensitivity to kernel choice and hyperparameter tuning. Additionally, it exhibited the highest latency (approximately 3.95 seconds), reflecting the known computational cost of SVR on large datasets.

4.7 Why These Results Are Expected

Football player valuation depends on complex and non-linear relationships between skill-based attributes, physical characteristics, and contract-related features. Ensemble tree-based models are well suited to capturing such interactions, which explains their superior performance.

Several engineered features, such as potential growth and remaining contract years, interact in ways that linear models cannot fully represent. In contrast, SVR struggled due to the high feature dimensionality, complex data structure, and sensitivity to parameter settings. Overall, the observed performance differences align well with theoretical expectations for each modelling approach.

Table 1: Model Performance Comparison

Model/Approach	MAE	RMSE	R^2	Adjusted R^2	Latency (sec)
Linear Regression	3.298553e+05	1.029650e+06	0.962521	0.962402	0.002092
Lasso Regression	3.298552e+05	1.029650e+06	0.962521	0.962402	0.004137
Decision Tree	1.712096e+05	6.691852e+05	0.984169	0.984119	0.003119
Random Forest	1.433107e+05	7.021547e+05	0.982571	0.982516	0.071678
Gradient Boosting	2.209176e+05	6.675338e+05	0.984247	0.984197	0.006780
Support Vector Regressor	1.884872e+06	5.546178e+06	-0.087415	-0.090870	3.951116

Key Visual Insights

Several visualizations were generated to understand relationships between features and the target variable:

- **Correlation Heatmap**: highlighted a cluster of strongly positive correlations among attributes such as overall rating, potential, release clause, and wage, confirming their relevance in value prediction.
- **Distribution Plots**: showed that the target variable is right-skewed, and log transformation helped stabilize variance and improve model performance.
- **Scatterplots**: revealed clear nonlinear relationships between player performance metrics and market value, supporting the higher performance of tree-based models.

Interesting Findings

- Player value is most strongly influenced by a combination of skill-based attributes, physical metrics, and contract-related features.
- PCA-based feature selection confirmed that variance in value can be explained using only a compact subset of influential variables.
- Tree-based models provided high accuracy with low inference time, making them suitable for real-time value estimation.
- Models like SVR are unsuitable for this dataset due to computational inefficiency and sensitivity to high dimensionality.

5 Discussion and Analysis

The results demonstrate that ensemble-based tree models are better suited to predicting football player market value than linear or kernel-based approaches. This outcome aligns with the inherently non-linear nature of player valuation, where attributes such as performance ratings, age, contract length, and potential interact in complex ways.

Random Forest and Gradient Boosting achieved the strongest predictive performance, though with slightly different error profiles. Random Forest produced lower average absolute errors, while Gradient Boosting reduced larger residuals more effectively. These differences are consistent with their underlying learning mechanisms and do not indicate a clear dominance of one ensemble method over the other.

Linear and Lasso Regression provided stable baseline performance but were constrained by their linear assumptions. Their identical results suggest that multicollinearity had already been addressed during feature selection, limiting the impact of regularization in the final dataset.

The Support Vector Regressor underperformed substantially and exhibited the highest latency. Given the high dimensionality introduced through encoding and the sensitivity of SVR to hyperparameter choices, this outcome highlights the limitations of kernel-based methods for large, structured tabular datasets in this context.

Latency served as a complementary metric rather than a primary evaluation criterion. While ensemble models required additional computation time, the increase was moderate and justified by improved accuracy. Linear models remained computationally efficient but less expressive.

Overall, the observed performance patterns are consistent with both the structure of the dataset and the theoretical properties of the models used.

6 Visualizations and Insights

This section presents key visualizations used to understand the dataset and interpret model performance. Graphs were used to examine the distribution of player market value, explore relationships between important features, and compare the performance of different machine learning models. These visual tools help explain the patterns observed in the data and support the modelling decisions made in the study. Each visualization is discussed individually to highlight its significance and the insights it provides.

6.1 Distribution of Player Market Value

The histogram of player market value shows a highly right-skewed distribution. Most players have relatively low market values, while a small number of elite players have extremely high values. This long right tail indicates the presence of outliers and large variation in prices across players. Such skewness can affect model performance, especially for models that assume a more balanced distribution of the target variable. This graph explains why raw market value is difficult to predict directly and highlights the need for transformation.

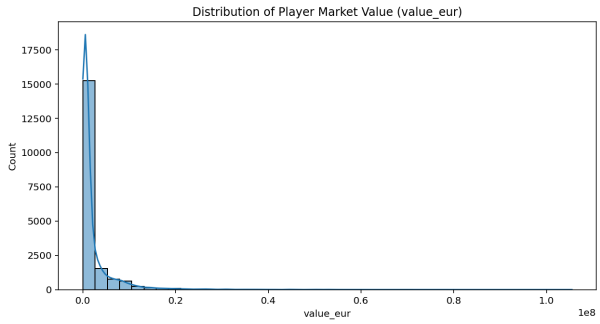

Figure 1: Distribution of Player Market Value

6.2 Distribution After Log Transformation of Market Value

After applying a logarithmic transformation to market value, the distribution becomes much more balanced and closer to a normal shape. Extreme values are compressed, and differences between low and mid-range values become clearer. This transformation reduces the influence of outliers and helps models learn patterns more effectively. The improved shape of the distribution supports the decision to use the log-transformed target during model training.

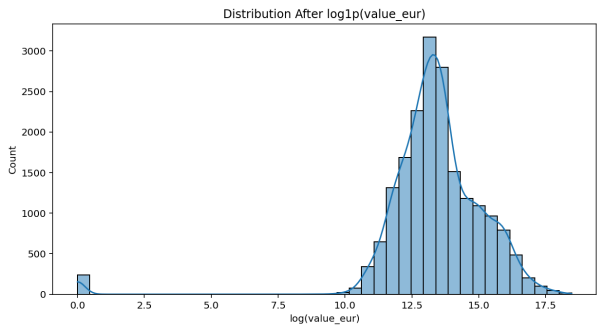

Figure 2: Distribution After Log Transformation of Market Value

6.3 Correlation Heatmap of Numerical Features

The correlation heatmap shows the relationships between numerical features and the target variable. Strong positive correlations are visible between market value and attributes such as overall rating, potential, and wage-related features. At the same time, many player attributes

are also highly correlated with each other, indicating multicollinearity. This visualization was important for identifying redundant variables and motivated the use of correlation-based filtering and PCA during feature selection.

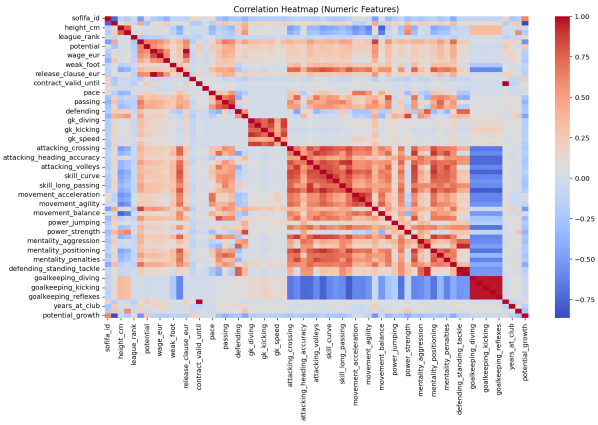

Figure 3: Correlation Heatmap of Numerical Features

6.4 Model Performance Comparison Plot

The model comparison plot presents MAE, RMSE, R^2, and Adjusted R^2 for all six models. It clearly shows that tree-based ensemble models, particularly Random Forest and Gradient Boosting, outperform linear models and Support Vector Regression across most metrics. Linear and Lasso Regression perform similarly, suggesting limited benefit from regularization after feature selection. The Support Vector Regressor shows poor performance, with high error values and low explanatory power. This visualization reinforces the conclusion that ensemble models are better suited for capturing the complex and non-linear patterns present in player valuation data.

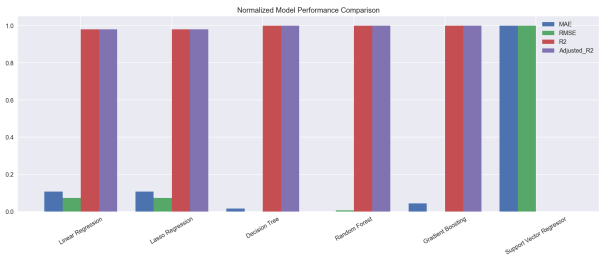

Figure 4: Model Performance Comparison Plot

6.5 Relationship Between Player Attributes and Market Value

The following graphs show the relationship between player market value and the attributes that have the highest correlation with it. Overall, most of the attributes show a clear positive relationship with market value, meaning that as the attribute score increases, the player's value also tends to increase. Attributes such as overall rating, potential, movement reactions, and international reputation show strong upward trends, indicating that better-rated and higher-potential players are generally valued more in the market. The scatter plots also show that market value increases more sharply at higher attribute levels, especially for top-performing players. The club-wise and position-wise distribution graphs highlight differences in player valuation across teams and roles. Players from top clubs usually have higher market values, while attackers and midfielders tend to be valued more compared to defenders and goalkeepers. Overall, these graphs suggest that player performance, future potential, reputation, and playing

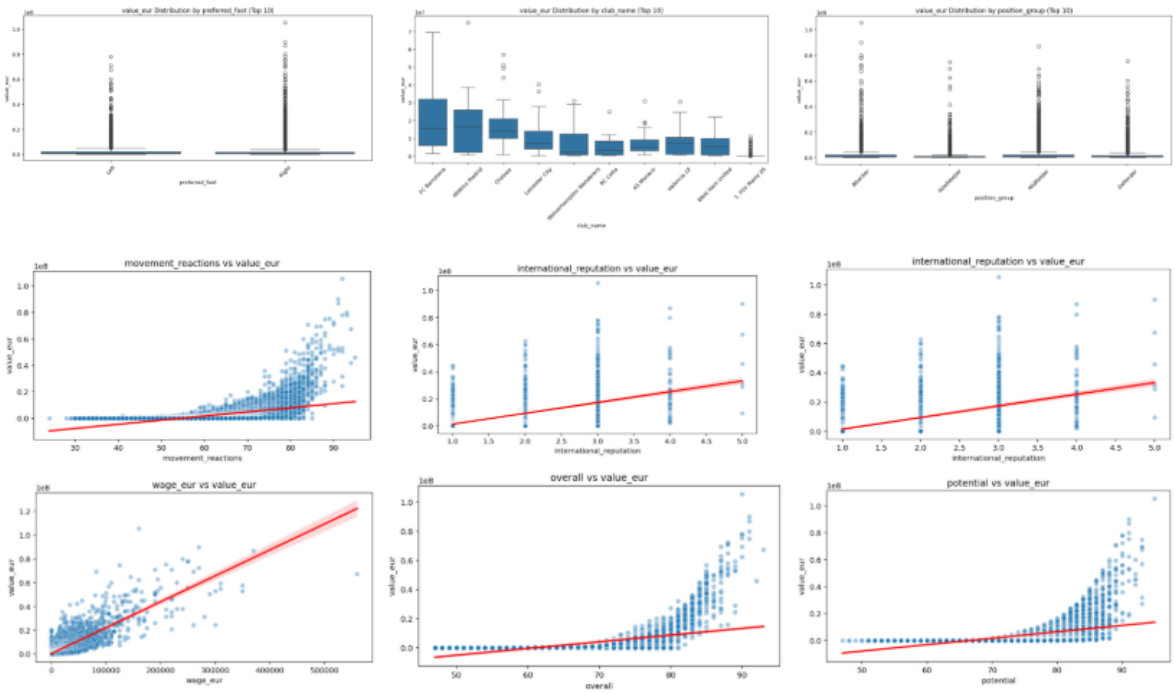

Figure 5: Relationship Between Player Attributes and Market Value

role all play an important role in determining market value, though the relationship is not perfectly linear and varies across players.

7 Conclusion

This project shows that proper data preprocessing and choosing suitable models are important when predicting football player market values. Comparing different models helped in understanding how each approach performs on the same dataset and what trade-offs exist between accuracy and computation time. Instead of focusing on one best model, the project highlights the value of evaluating multiple methods to better understand their strengths and limitations. Overall, the project successfully meets its goal of comparing regression models in a practical sports analytics context.

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Individual Contributions

- **Anshi Doshi**: Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Handling Missing Values, Feature Selection, Model Implementation, Hyperparameter Tuning, Model Evaluation, Visualization and Insight Generation, Code Organization, Report Writing

- **Prakarti**: Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Handling Missing Values, Feature Selection, Model Implementation, Hyperparameter Tuning, Model Evaluation, Visualization and Insight Generation, Code Organization, Report Writing